MATH256
Linear Algebra HW#3

1. (2.1.12) Let A = $\begin{bmatrix} 3 & -6 \\ -2 & 4 \end{bmatrix}$. Construct a 2 x 2 matrix B such that AB is the zero matrix. Use two different nonzero columns for B.
2. (2.1.15) A , B , and C for which the indicated sums and products are defined. Mark each statement True or False. Justify each answer.
   a. If A and B are 2 x 2 matrices with columns $\boldsymbol{a_1}$, $\boldsymbol{a_2}$ and $\boldsymbol{b_1}$ $\boldsymbol{b_2}$ respectively, then AB = [$\boldsymbol{a_1 b_1}$ $\boldsymbol{a_2 b_2}$] .
   b. Each column of AB is a linear combination of the columns of B using weights from the corresponding column of A.
   c. AB + AC = A(B + C)
   d. $A^T + B^T = (A + B)^T$
   e. The transpose of a product of matrices equals the product of their transposes in the same order.
3. (2.1.16) A, B, and C for which the indicated sums and products are defined. Mark each statement True or False .Justify each answer.
   a. The first row of AB is the first row of A multiplied on the right by B.
   b. If A and B are 3 x 3 matrices and B =[$\boldsymbol{b_1}$ $\boldsymbol{b_2}$ $\boldsymbol{b_3}$], then AB= [$\boldsymbol{Ab_1 + Ab_2 + Ab_3}$]
   c. If A is an n x n matrix ,then $(A^2)^T = (A^T)^2$
   d. $(ABC)^T = C^T A^T B^T$
   e. The transpose of a sum of matrices equals the sum of their transposes.
4. (2.1.26) Suppose AD = $\boldsymbol{I_m}$ (the m x m identity matrix). Show that for any **b** in $\mathbb{R}^m$, the equation **Ax = b** has a solution. [Hint : Think about the equation Ad**b=b**] Explain why A cannot have more rows than columns.

5. (2.2.9) Mark each statement True or False. Justify each answer.

a. In order for a matrix B to be the inverse of A, the equation AB = I and BA = I must both be true T

b. If A and B are n x n and invertible, then $A^{-1}B^{-1}$ is the inverse of AB. F B^-1 A^-1

c. If A = $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and ab – cd ≠ 0, then A is invertible.F ad - bc != 0

d. If A is invertible n x n matrix, then the equation **Ax = b** is consistent for each b in $\mathbb{R}^n$. T

e. Each elementary matrix is invertible. T

6. (2.2.10) Mark each statement True or False .Justify each answer.

a. If A is invertible, then elementary row operations that reduce A to the identity $\boldsymbol{I_n}$ also reduce A^{-1} to I_n . F I to A^-1

b. If A is invertible , then the inverse of A^{-1} is A itself T

c. A product of invertible n x n matrices is invertible, and the inverse of the product is the product of theirs inverses in the same order. F

d. If A is an n x n matrix and **Ax =** e_j is consisted for every j ε { 1,2,. . . . , n }, then A is invertible. Note: e_1 , . . . , e_n represented the columns of identity matrix . T

e. If A can be row reduced to the identity matrix, then A must be invertible. T

7. (2.2.32) Find the inverses of the matrices in this matrix, if it exists. Use algorithm introduced in this section.

$$\begin{bmatrix} 1 & 2 & -1 \\ -4 & -7 & 3 \\ -2 & -6 & 4 \end{bmatrix}$$

8. (2.3.6) Unless otherwise specified, assume that all matrices in these questions are n x n. Determine which of the matrices in this questions (8-9) are invertible. Use as few calculations as possible justify your answer.

$$\begin{bmatrix} 1 & -3 & -6 \\ 0 & 4 & 3 \\ -3 & 6 & 0 \end{bmatrix}$$

9. (2.3.8) Unless otherwise specified , assume that all matrices in these questions are n x n . Determine which of the matrices in this questions (8-9) are invertible. Use as few calculations as possible justify your answer.

$$\begin{bmatrix} 3 & 4 & 7 & 4 \\ 0 & 1 & 4 & 6 \\ 0 & 0 & 2 & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

In questions 10 and 11, the matrices are all n x n . Each part of this questions is an implication of form "If <statement 1 > then <statement 2 >." Mark an implication as True if the truth of <statement 2 > always follows whenever <statement 1 > happens to be true. An implication is False if there is an instance in which <statement 2> is false but <statement 1> is true. Justify each answer.

10. (2.3.11)
    a. If the equation **Ax=0** has only the trivial solution, then A is row equivalent to the n x n identity matrix. T
    b. If the columns of A span $\mathbb{R}^n$, then the columns are linearly independent. T
    c. If A is an n x n matrix , then the equation **Ax=b** has at least one solution for each **b** in $\mathbb{R}^n$ T
    d. If the equation **Ax= 0** has a nontrivial solution, then A has fewer than n pivot positions. T
    e. If A^T is not invertible, then A is not invertible. T

11. (2.3.12)
    a. If there is an n x n matrix D such that AD = 1, then DA = 1 . F
    b. If the linear transformation **x |-> Ax** maps $\mathbb{R}^n$ into $\mathbb{R}^n$, then the row reduced echelon form of A is I . T
    c. If the columns of A are linearly independent, then the columns of A span $\mathbb{R}^n$. T
    d. If the equation **Ax = b** has at least one solution for each **b** in $\mathbb{R}^n$, then the transformation **x |-> Ax** is not one-to-one F
    e. If there is a **b** in $\mathbb{R}^n$ such that the equation **Ax = b** is consistent, then the solution is unique.
12. (2.3.14) An m x n **lower triangular matrix** is one whose entries above the main diagonal are 0's (as in exercise 2.3.3 in book). When is a square lower triangular matrix invertible? Justify your answer.
13. (2.5.6) Solve the equation Ax= b by using the LU factorization given for A.

$$A=\begin{bmatrix} 1 & 3 & 2 & 0 \\ -2 & -3 & -4 & 12 \\ 3 & 0 & 4 & -36 \\ -5 & -3 & -8 & 49 \end{bmatrix}, b=\begin{bmatrix} 1 \\ -2 \\ -1 \\ 2 \end{bmatrix}$$

$$A=\begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 3 & -3 & 1 & 0 \\ -5 & 4 & -1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 3 & 0 & 12 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Find the LU factorization of the matrices in **Questions 14-15** (with L unit lower triangular). Note that MATLAB will usually produce a permuted LU factorization because it uses partial pivoting for numerical accuracy.

14. (2.5.14)
$$\begin{bmatrix} 1 & 3 & 1 & 5 \\ 5 & 20 & 6 & 31 \\ -2 & -1 & -1 & -4 \\ -1 & 7 & 1 & 7 \end{bmatrix}$$

15. (2.5.16)
$$\begin{bmatrix} 2 & -3 & 4 \\ -4 & 8 & -7 \\ 6 & -5 & 14 \\ -6 & 9 & -12 \\ 8 & -6 & 19 \end{bmatrix}$$